

Zhangir Nurmukhambetov

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TECHNICAL SKILLS

Languages: C++, C#, GLSL, HLSL, Python

Frameworks & Libraries: OpenGL, EnTT (ECS), ImGui, Cereal, GLM, tinyglTF, ImGuizmo

Game Engines: Unreal Engine 5 (Blueprints & C++), Unity (C#), Godot

Developer Tools: Visual Studio, Visual Studio Code, Git, Perforce (P4V), GitHub Actions

SELECTED PROJECTS

Zentera | *Custom C++ Engine, Angelscript, FMOD*

May 2025 – Jun 2025

- Served as Programming Representative Lead, acting as the primary point of contact for programming tasks and coordinating communication between programmers, artists, designers, and external stakeholders.
- Developed a fully featured **Particle System** in C++ using EnTT ECS, modeled after Unity's workflow, including shape modules, color gradients, and object pooling for performance optimization.
- Integrated FMOD **Sound Events** into gameplay scripts, synchronizing audio cues for interactions like player footsteps, tongue actions, crates, levers, and environment elements.
- Reported and fixed critical bugs impacting gameplay stability, visuals, and performance, including collision errors, particle-induced lag, and level design oversights.
- Led the **release process** on itch.io: coordinated page creation, added team members as contributors, curated marketing assets, defined minimum hardware specifications, and contributed custom trailer scenes.
- Collaborated across disciplines, ensuring engine features and visual/audio feedback were aligned with design goals, resulting in a polished early access release and multiple updates informed by player feedback.

Fire Ant Engine | *C++, OpenGL, GLSL, EnTT, ImGui, Cereal*

Feb 2025 – Apr 2025

- Led development of a custom C++ strategy game engine featuring a terrain editor, prefab system, flow-field AI navigation, and in-engine UI editor.
- Engineered terrain rendering and editing tools, including height and texture manipulation, prop placement, and scene serialization.
- Optimized engine performance to support large maps and numerous units on Windows and Nintendo Switch platforms.
- Delivered a fully functional strategy game engine within an 8-week timeframe, laying the groundwork for future projects and cross-platform compatibility.

Custom Terrain Editor | *C++, OpenGL, GLSL, ImGui, GLM*

Nov 2024 – Jan 2025

- Developed a solo research project focusing on editable terrain with various brush tools, serving as the foundation for the Fire Ant Engine.
- Implemented terrain mesh deformation via height maps, dynamic normals, and displacement using vertex shaders.
- Established a robust terrain editing toolset that was directly integrated into the Fire Ant Engine, enhancing its level design capabilities.

Custom C++ Game Engine | *C++, EnTT, ImGui, Cereal, STB, tinyglTF, ImGuizmo*

Sep 2024 – Nov 2024

- Designed and implemented core engine functionalities including ECS architecture, particle system, ImGui editor with gizmo tooling, and GLTF runtime loader.
- Developed a resource manager, hierarchy system, cross-platform engine abstractions, JSON serialization, and a tile-based level editor.
- Delivered a versatile and extensible game engine over 8 weeks, providing a solid foundation for subsequent projects and cross-platform development.

IgKnighited | *Unreal Engine 5*

May 2024 – Jun 2024

- Developed core gameplay systems for a twin-stick shooter, including input mapping, local multiplayer, enemy attacks, player damage, collectables, power-ups, and in-engine animations.
- Collaborated with a multidisciplinary team of 4 programmers, 3 designers, and 5 artists to deliver a cohesive gaming experience on Windows.
- Successfully launched the game on itch.io, receiving positive feedback and facilitating further development through community engagement.

EDUCATION

Breda University of Applied Sciences

Bachelor of Science in Creative Media and Game Technologies

September 2023 – Present

Breda, Netherlands